

Hall Ticket No:

--	--	--	--	--	--	--	--	--	--

Course Code: A8604

**VARDHAMAN**

COLLEGE OF ENGINEERING

III B.Tech I Semester Continuous Assessment – II, October- 2025

(Regulations: VCE-R22)

WEB TECHNOLOGIES**(Information Technology & Computer Science and Engineering)**

Date: 23 October 2025 FN

Time: 2 Hours

Max. Marks: 30

Answer all Questions in Part-A
Answer any FOUR Questions in Part-B

Course Outcomes with Bloom's Levels:

CO#	CO Statement	Bloom's Level (L#)
CO1	Design static and interactive web pages for a given application.	L3
CO2	Choose a web standard to store, transport and exchange in heterogeneous environments.	L3
CO3	Make use of JDBC API to interact with various databases.	L3
CO4	Deploy server-side components for providing services to the client.	L3
CO5	Design dynamic and data driven web pages in client-server environment.	L3

Questions:

PART-A					
			CO#	BL#	Marks
1.	a)	Analyze the purpose of the Class.forName() method in JDBC and how it impacts database connectivity.	CO3	L3	1 M
	b)	Differentiate between the various JDBC interfaces that can be used to execute SQL queries, and identify the most suitable one for different scenarios.	CO3	L3	1 M
	c)	Explain the role of the ResultSet interface in JDBC and how it manages data retrieval from a database.	CO3	L2	1 M
	d)	Demonstrate the usage of Servlet interfaces that handle HTTP requests and responses, focusing on their roles in a web application.	CO4	L3	1 M
	e)	Explain the purpose of the destroy() method in Servlets, and analyze when it should be utilized in the servlet lifecycle.	CO4	L2	1 M
	f)	Apply your understanding of Servlets to describe the method used to send a cookie to the client.	CO4	L3	1 M
	g)	Examine the JSP tag that is used to declare a Java method, and explain how it integrates Java logic within a JSP page.	CO5	L3	1 M
	h)	Explain the purpose of the include directive in JSP, and differentiate it from other JSP directives and actions.	CO5	L2	1 M

	i)	Demonstrate the method in JSP that is used to delete a cookie, explaining its practical application in a web context.	CO5	L3	1 M
	j)	Analyze the types of JDBC drivers available, focusing specifically on the one used for database-specific drivers, and justify its use in various situations.	CO4	L3	1 M
PART-B					
2.		Outline the steps involved in establishing a database connection using JDBC, analyzing how each step contributes to the overall connection process.	CO3	L3	5 M
3.		Analyze the concept of a GIT repository by examining its components and structure, and evaluate how the process of creating a repository supports efficient version control and collaboration.	CO3	L4	5 M
4.		Illustrate the concept of Cookies with an example, analyzing their role in web development and data management.	CO4	L3	5 M
5.		Examine the lifecycle of a servlet, breaking down each phase with examples of what occurs during each step.	CO4	L4	5 M
6.		Use the concept of JSP implicit objects, identifying their uses and how they streamline web application development.	CO5	L3	5 M
7.		Analyze how various standard action elements in JSP differ in their roles and functionality, and examine how each contributes to performing specific tasks within a JSP page	CO5	L4	5 M



VARDHAMAN

COLLEGE OF ENGINEERING
(AUTONOMOUS)

(Affiliated to JNTUH, Approved by AICTE and Accredited by NBA)
Shamshabad, Hyderabad - 501 218

SCHEMA OF VALUATION

Name of the Subject : Web Technologies
Programme : IT
Examination : Regular / Supplementary
Max. Marks : 30

Subject Code : A8604
Year : III
Date of Examination : 23/10/15
Time of Examination : FN

- 7
- a) class.forName() loads the JDBC driver class dynamically at runtime, registering with the DriverManager.
- 0.5M
- 0.5M Syntax class.forName("com.mysql.cj.jdbc.Driver");
- b)
- Statement - for simple static SQL queries
- Prepared Statement - for precompiled, parameterized queries.
- 0.5M
- Callable Statement - for executing stored procedures
- 0.5M
- Most suitable : prepared statement for dynamic & frequent execution.
- c)
- ResultSet represents the result of a SQL query. It provides cursor based navigation to retrieve, update, access data row by row using getter methods.
- 1M

14) } d) Servlets implement `HttpServletRequest` and `HttpServletResponse` to handle client requests and generate dynamic responses

14) } e) `destroy()` method is called once before the servlet is unloaded. It's used to release resources such as database connections.

14) } f) `response.addCookie(cookie c);`

14) } g) `<%! %>` declaration tag

14) } h) `<%@ include file = "index.jsp" %>` includes static content at transition time
`<jsp:include>` which includes dynamically at request time.

14) } i) To delete a cookie, set its max age to zero
`Cookie.setMaxAge(0);`
`response.addCookie(cookie);`

14) } j) 1) JDBC-ODBC Bridge
 2) Native API
 3) Network Protocol
 4) Thin Driver — database specific platform independent & most efficient for production use

PART- B

- 2M {
1. Import JDBC packages
 2. Load the Driver
 3. Establish Connection.
 4. Create a stmt Object
 - 5) Execute Queries
 - 6) close the Connection

- 3M {
- 1) `import java.sql.*`
 - 2) `class.forName("com.mysql.cj.jdbc.Driver")`
 - 3) `Connection con = DriverManager.getConnection`
`(url, user, pwd)`
 - 4) `Statement stmt = con.createStatement()`
 - 5) `ResultSet rs = stmt.executeQuery ("select`
`* from table)`
 - 6) `con.close()`

3) Git Components :-

- 2M {
- 1) git directory - Internal Storage containing commits, branches
 - 2) Working Directory - Contains the current version of project files
 - 3) staging area - Prepares files for commit
 - 4) commit - Snapshot of the project
Guide
 - 5) Branches - Independent lines of development for parallel work.



Git process

- 3M {
- 1) git init
 - 2) git add
 - 3) git commit
 - 4) git remote add origin URL
 - 5) git push.
- } with examples

4) A cookie is a small text file stored on the client's browser to maintain stateful information b/w client & server

1M { Ex:- `Cookie c1 = new Cookie("name", "x");`
`response.addCookie(c1);`

To retrieve

2M { `Cookie[] CK = request.getCookies();`
`for (Cookie c : CK) {`
`if (c.getName().equals("uname")) {`
`out.println(c.getValue());`

Role in web Development :-

1M {

- 1) session management
- 2) tracking

5) Servlet lifecycle phases:-

1) Loading & Instantiation

2) Initialization

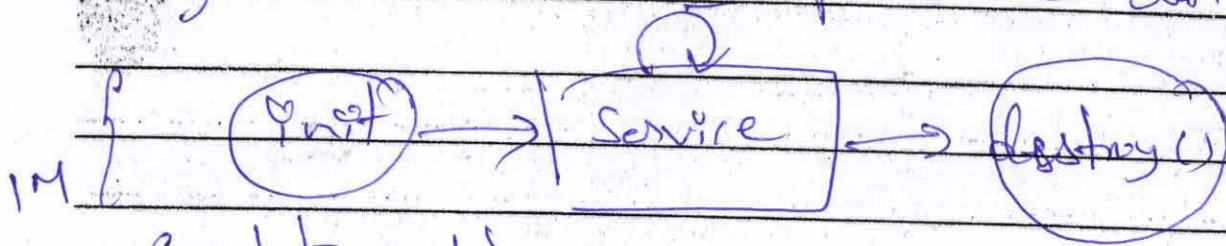
→ public void init()

3) Request handling

→ public void service()

4) Destruction

→ public void destroy()



Servlet skeleton:-

```
import java.io.*
```

```
import java.servlet.*
```

```
public class Servlet extends GenericServlet
```

```
{  
    public void init() throws ServletException
```

```
{  
    public void service(ServletRequest req,  
        ServletResponse res) throws ServletException,  
        IOException
```

```
{  
    public void destroy()
```



6) 9 Implicit Objects in Jsp :-

Object → Access client data
 request → send response to client
 response

Session → Manage user sessions

appln → share data across pages

out → send output to client

Config → Access Servlet Configuration

PageContext → Access page level data

page → Refers to a current Jsp

Exception → page.

to handle Exception.

SM
with
any
files
Event

7)

① `<jsp:include>` used to include another jsp file

Bas - `<jsp:include page="url" />`

② `<jsp:forward>` used to forward the request to another resource such as html ; Jsp page

③ `<jsp:plugin>`

used to insert java components in a Jsp page

Syntax `<jsp:plugin type=" " align=" " />`

for
any
with
SM

(4) `<jsp:useBean>`

→ it is used to find or instantiate a JavaBean.

Syntax `<jsp:useBean id="x", class="one.class">`

(5) `<jsp:setProperty>`

(6) `<jsp:getProperty>`





A series of horizontal lines spanning the width of the page, intended for writing. There are approximately 25 lines in total, evenly spaced.



A blank sheet of lined paper with horizontal ruling lines.



VARDHAMAN

COLLEGE OF ENGINEERING
(AUTONOMOUS)

(Affiliated to JNTUH, Approved by AICTE and Accredited by NBA)
Shamshabad, Hyderabad – 501 218

EXAMINATION BRANCH

Instructor's Feedback on Question Paper

A8604

The report has been filed even if there is no adverse feedback on the question paper

Programme : <u>IT/CSE</u>	Year: <u>III</u>	Semester: <u>I</u>
Course Title: <u>Web Technologies</u>		
Date of Examination: <u>23/10/25</u>	Time of Examination: <u>PM</u>	Max. Marks: <u>30</u>
Course Coordinator: <u>B. Swapna</u>	Course Instructor(s) 1) <u>K. Anvesh</u> 2) <u>G. S. Parasada Reddy</u> 3) 4) 	

Report on Question Paper	
1. Whether the question paper is as per the pattern briefed to the students? Tick appropriately. Yes ☒ No ☐ [If the answer is Yes, go to Item 3]	
2. If the answer to item 1 above is No, mention the deviation in question paper vis-à-vis the prescribed pattern. (If required, use additional paper) 	
3. Comment on the sufficiency of prescribed time to answer the questions. Tick appropriately. Just sufficient ☒ Time is in excess ☐ Time is not sufficient ☐	
4. Are the questions distributed among various modules / units of the course as per the prescribed pattern? Tick appropriately. Yes ☒ No ☐ [if the answer is Yes, go to Item 6]	
5. If the answer to Item 4 above is No, provide full details on the skewness in question paper? (If required, use additional paper) 	
6. Are there any exceptions / anomalies in question paper? Tick appropriately. Yes ☐ No ☒ [If the answer is No, go to item 8]	

7. If the answer to item 6 above is Yes, provide details of exceptions / anomalies in the question paper along with remedial actions suggested for handling them. Examination department may consider these suggestions, if deem fit.

Unit	Q. No.	Marks	Exception / Anomaly	Remedial Action Suggested

8. Are the questions well balanced between concepts, theoretical questions and application / problem based / case study questions? Tick appropriately.

Balanced ☒

Theory oriented ☐

Application oriented ☐

9. How do you rate the overall quality of the question paper? Tick appropriately.

Excellent ☒

Very Good ☐

Good ☐

Satisfactory ☐

Unsatisfactory ☐

10. Your suggestions for improvement in quality of question paper in future.

Nil

1. Name of the Instructor: B. Anwar

Signature [Signature]

2. Name of the Instructor: _____

Signature _____

3. Name of the Instructor: _____

Signature _____

Signature of HOD [Signature]

Signature of Course Coordinator [Signature]

Date and Time: 24/10/2015

Date and Time: 24/10/15

REMARKS OF THE HEAD OF THE DEPARTMENT

Date and Time: 24/10/25

Signature of Controller of Examinations [Signature]